

authorities started getting offended. Likewise we think that Galileo was a victim in the name of science; in fact, the church didn't take him too seriously. It seems, rather, that Galileo caused the uproar himself by ruffling a few feathers. At the end of the year in which Darwin and Wallace presented their papers on evolution by natural selection that changed the way we view the world, the president of the Linnean society, where the papers were presented, announced that the society saw "no striking discovery," nothing in particular that could revolutionize science.

We forget about unpredictability when it is our turn to predict. This is why people can read this chapter and similar accounts, agree entirely with them, yet fail to heed their arguments when thinking about the future.

Take this dramatic example of a serendipitous discovery. Alexander Fleming was cleaning up his laboratory when he found that penicillium mold had contaminated one of his old experiments. He thus happened upon the antibacterial properties of penicillin, the reason many of us are alive today (including, as I said in [Chapter 8](#), myself, for typhoid fever is often fatal when untreated). True, Fleming was looking for "something," but the actual discovery was simply serendipitous. Furthermore, while in hindsight the discovery appears momentous, it took a very long time for health officials to realize the importance of what they had on their hands. Even Fleming lost faith in the idea before it was subsequently revived.

In 1965 two radio astronomers at Bell Labs in New Jersey who were mounting a large antenna were bothered by a background noise, a hiss, like the static that you hear when you have bad reception. The noise could not be eradicated—even after they cleaned the bird excrement out of the dish, since they were convinced that bird poop was behind the noise. It took a while for them to figure out that what they were hearing was the trace of the birth of the universe, the cosmic background microwave radiation. This discovery revived the big bang theory, a languishing idea that was posited by earlier researchers. I found the following comments on Bell Labs' website commenting on how this "discovery" was one of the century's greatest advances:

Dan Stanzione, then Bell Labs president and Lucent's chief operating officer when Penzias [one of the radio astronomers involved in the discovery] retired, said Penzias "embodies the creativity and technical excellence that are the hallmarks of Bell Labs." He called him a Renaissance figure who "extended our fragile understanding of creation, and advanced the frontiers of science in many important areas."

Renaissance shmenaissance. The two fellows were looking for bird poop! Not only were they not looking for anything remotely like the evidence of the big bang but, as usual in these cases, they did not immediately see the importance of their find. Sadly, the physicist Ralph Alpher, the person who initially conceived of the idea, in a paper coauthored with heavyweights George Gamow and Hans Bethe, was surprised to read about the discovery in